26900072 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBOT304, MG5DSEBBT302 - ANALYTICAL TECHNIQUES IN PLANT SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. What is glycerine and what is its use in microtechnique? [U] / [CO1]
2. List out the difference between sledge and rotary microtome. [K] / [CO1]
3. Define the smear technique used in plant cytological studies. [K] / [CO1]
4. Define maceration in plant microtechnique. [K] / [CO1]
5. State the two basic methods used for preserving plant specimens. [K] / [CO1]
6. Name different types of fixation. [K] / [CO1]
7. What is meant by chamber volume in haemocytometer counting? [K] / [CO4]
8. Mention any two applications of UV-Visible spectroscopy. [U] / [CO5]
9. How does particle size affect sedimentation? [A] / [CO4]
10. Describe the principle of Lyophilizer. [U] / [CO4]

11. Which electrode do DNA molecules migrate toward during agarose gel electrophoresis? [U] / [CO2]

12. What are the different components of TLC? [U] / [CO5]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **8** questions

Each question carries **5** marks

13. Explain the preparation and use of safranin in plant microtechnique. [U] / [CO1]

14. Describe the general classification of fixatives with suitable examples [U] / [CO1]

15. Explain the working of a compound light microscope. [U] / [CO3]

16. Explain the principle and working of a spectrophotometer. [U] / [CO2]

17. Explain the structural components of a pH meter. [U] / [CO3]

18. Describe the process of image formation in SEM. [U] / [CO3]

19. Explain the principle and applications of ultracentrifugation. [U] / [CO2]

20. Describe how the electric charge of a molecule affects electrophoretic mobility. [U] / [CO2]

21. Briefly explain the working and application of paper chromatography. [U] / [CO3]

22. Compare and contrast HPLC and column chromatography. [U] / [CO5]

[5x8 = 40]

Part C

Essay Questions

Answer any **1** questions

Each question carries **10** marks

23. Distinguish between temporary and permanent slide preparation techniques. Discuss the steps involved and suitable mounting media for each. [U] / [CO1]

24. Explain the working and applications of colorimeter. [U] / [CO3]

[10x1 = 10]